“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 21

Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.2 (line integrals)

- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

ALVARO: Start the recording!

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The Fundamental Theorem for Line Integrals

The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus, Part 2

If f is continuous on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

where F is any antiderivative of f, that is, a function F such that $F' = f$.

$$\int_a^b F'(x)\,dx = F(b) - F(a)$$

The Fundamental Theorem for Line Integrals

2 Theorem

Let C be a smooth curve given by the vector function $\mathbf{r}(t)$, $a \leqslant t \leqslant b$. Let f be a differentiable function of two or three variables whose gradient vector ∇f is continuous on C. Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

The Fundamental Theorem for Line Integrals: Proof

The Fundamental Theorem for Line Integrals: Proof

$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$

DOT

$r'(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$

$$\int_C \nabla f \cdot d\mathbf{r} = \int_a^b \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt$$

$$= \int_a^b \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) dt$$

$$= \int_a^b \frac{d}{dt} f(\mathbf{r}(t))\, dt \qquad \text{(by the Chain Rule)}$$

$$= f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

Example: Let $f(x,y) = x^2 e^y$. Evaluate $\int_C \nabla f \cdot dr$ where C is the curve parametrized by $r(t) = \left(2t, t^3\right)$, $1 \le t \le 2$.

$$\nabla f = \left(2xe^y,\ x^2 e^y\right) = \left(4te^{t^3},\ 4t^2 e^{t^3}\right)$$

$$r'(t) = \left(2,\ 3t^2\right)$$

$$\nabla f \cdot r'(t) = 8te^{t^3} + 12t^4 e^{t^3}$$

$$\int_1^2 8te^{t^3} + 12t^4 e^{t^3}\,dt$$

[Extra Space]

Example: Let $f(x,y) = x^2 e^y$. Evaluate $\int_C \nabla f \cdot dr$ where C is the curve parametrized by $r(t) = \left(2t, t^3\right)$, $1 \le t \le 2$.

$$\int_C \nabla f \cdot dr = f(r(2)) - f(r(1))$$

$$= \boxed{16e^8 - 4e}$$

$$r(1) = (2,1)$$

$$r(2) = (4,8)$$

$$f((2,1)) = 2^2 \cdot e^1 = 4e$$

$$f((4,8)) = 4^2 e^8 = 16e^8$$

Example: Let $F(x, y, z) = (yz, xz, xy)$. Evaluate $\int_C F \cdot dr$ where C is the line segment from (1,0,-2) to (4,6,3).

Q Is $F = \nabla f$, for some f??

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (yz, xz, xy)$$

GUESS $\underbrace{f = xyz}_{\text{potential!}}$

$$\int_C F \cdot dr = \int_C \nabla f \cdot dr = f(r(b)) - f(r(a))$$

[Extra Space]

Example: Let $F(x,y,z) = (yz, xz, xy)$. Evaluate $\int_C F \cdot dr$ where C is the line segment from (1,0,-2) to (4,6,3).

FT of LINE INTEGRALS!

$$\int_C F \cdot dr = \int_C \nabla f \cdot dr = f\left(\begin{smallmatrix}\text{end}\\ \text{pt}\end{smallmatrix}\right) - f\left(\begin{smallmatrix}\text{start}\\ \text{pt}\end{smallmatrix}\right)$$

$$f(x,y,z) = xyz$$

$$f(4,6,3) = 4 \cdot 6 \cdot 3 = 72$$

$$f(1,0,-2) = 1 \cdot 0 \cdot (-2) = 0$$

$$= 72 - 0$$

$$= \boxed{72}$$

Conservative Vector Fields

A vector field $\mathbf{F}$ is called a **conservative vector field** if it is the gradient of some scalar function, that is, if there exists a function f such that $\mathbf{F} = \nabla f$. In this situation f is called a **potential function** for $\mathbf{F}$.

Example: Let $F(x, y, z) = (yz, xz, xy)$. Is F conservative?

$$f(x,y,z) = xyz \Rightarrow \nabla f = F$$

Example: Let $G(x, y, z) = (xy^2 + 3x, x^2y)$. Is G conservative? ???

WORKS!

$$\frac{\partial f}{\partial x} = xy^2 + 3x \to f = \int xy^2 + 3x \, dx = \underbrace{\boxed{\frac{1}{2}x^2y^2 + \frac{3}{2}x^2}} + C(y)$$

$$\frac{\partial f}{\partial y} = x^2y + 0 + C'(y) = x^2y$$

YES

Example: Let G$(x, y, z) = (xy^2 + 3x, x^2y)$. Is G conservative?

If $G = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$

If so $\frac{\partial f}{\partial y\, \partial x} = \frac{\partial f}{\partial x\, \partial y}$ (Clairaut's Theorem)

$$\frac{\partial}{\partial y}(xy^2 + 3x) = 2xy$$

$$\|$$

$$\frac{\partial}{\partial x}(x^2y) = 2xy$$

f exists, G is conservative!

[Extra Space]

Example: Let $G(x,y,z) = (xy^2 + 3x, x^2y)$. Is G conservative?

1) Find f s.t. $\nabla f = G$

a) $\int xy^2 + 3x \, dx = \frac{1}{2}x^2y^2 + \frac{3}{2}x^2 + C(y)$

b) $\frac{\partial}{\partial y}\left(\frac{1}{2}x^2y^2 + \frac{3}{2}x^2\right) = x^2y + C'(y) = x^2y$

$\Rightarrow C'(y) = 0 \Rightarrow C(y)$ a const $= 0$

c) $\boxed{f = \frac{1}{2}x^2y^2 + \frac{3}{2}x^2}$

Recall!

Example: Let $F(\vec{x}) = -\frac{mMG}{|\vec{x}|^3}\vec{x}$ be a gravitational vector field (in 3D), and let $f(\vec{x}) = \frac{mMG}{|\vec{x}|}$. Show that $F = \nabla f$.

$$\int_C F(r(t)) \cdot dr = f(r(b)) - f(r(a))$$

Example: Evaluate $\int_C 2xdx + 2ydy$ where C is the curve parametrized by $r(t) = \left(t^2 - 1, \sqrt{t+1}\right),\ 0 \le t \le 3.$

$$\int_C 2x\,dx + 2y\,dy = \int_C (2x, 2y) \cdot dr$$

$$= 68 - 2 = \boxed{66}$$

FTOLI

$$F = (2x, 2y) = \nabla f$$

$$f = x^2 + y^2$$

$$r(0) = \left(0-1, \sqrt{0+1}\right) = (-1, 1) \qquad f(-1,1) = 2$$

$$r(3) = (8, 2) \qquad f(8,2) = 8^2 + 2^2 = 68$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P\,dx + Q\,dy + R\,dz \qquad \text{where } \mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k}$$

[Extra Space]

Example: Evaluate $\int_C 2xdx + 2ydy$ where C is the curve parametrized by

$$r(t) = \left(t^2 - 1, \sqrt{t+1}\right),\ 0 \le t \le 3.$$

$$F = (2x, 2y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$$

$$\frac{\partial}{\partial y}\left(\int 2x\, dx = x^2 + C(y)\right) = 0 + C'(y) = 2y$$

$$C(y) = \int C'(y) = 2y\, dy = y^2 + k$$

$$\boxed{f = x^2 + y^2 + k}$$

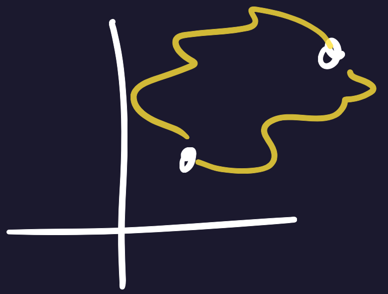
Path Independence

The line integrals of a vector field F are called path-independent if for any two paths C, D starting and ending at the same point we have

$$\int_C F \cdot dr = \int_D F \cdot dr.$$

Note: If F is conservative, then it is path-independent, by the Fundamental Theorem of Line Integrals.

$$\int_C F \cdot dr = \int_C \nabla f \, dr = f(\text{end}) - f(\text{start})$$



Example: Let $G(x, y, z) = (xy^2 + 3x, x^2y)$. Then G is path-independent.

$$G = \nabla\left(\frac{1}{2}x^2y^2 + \frac{3}{2}x^2\right) \Rightarrow \text{cons.} \Rightarrow \text{path-ind!}$$

Path Independence

The line integrals of a vector field F are called path-independent if for any two paths C_1, C_2 starting and ending at the same point we have

$$\int_{C_1} F \cdot dr = \int_{C_2} F \cdot dr.$$

A parametrized curve is called closed or a loop if it starts and ends at the same point.

$F = \nabla f$, C loop

$\int_C F\,dr = \int_C \nabla f\,dr = f(\text{end}) - f(\text{st}) = 0$

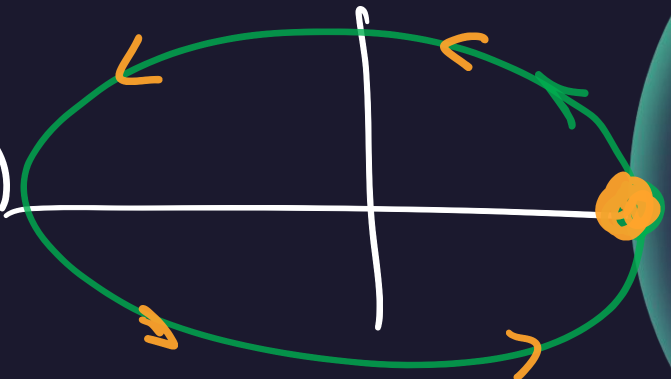
Example: Let $r(t) = (\cos(t), \sin(t)),\ 0 \le t \le 2\pi.$
Then r(t) is a loop.

3 Theorem

$\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in D if and only if $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path C in D.

Example: Evaluate $\int_C F \cdot dr$ where C is the ellipse $4x^2 + 9y^2 = 1$ and
$\mathrm{F} = \left((1+xy)e^{xy}, x^2 e^{xy}\right)$.

$$\int x^2 e^{xy}\, dy = x e^{xy} + C_1(x)$$



$$\frac{d}{dx}\left(x e^{xy} + C(x)\right) = e^{xy} + xy e^{xy} + C'(x)$$

$$\overset{?}{=} (1+xy)e^{xy} = e^{xy} + xy e^{xy}$$

$$\Rightarrow C'(x) = 0 \Rightarrow C(x) = 0$$

works.

$$\boxed{f = x e^{xy}} \qquad \nabla f = F$$

conservative!!

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C P\,dx + Q\,dy + R\,dz \qquad \text{where } \mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k}$$

[Extra Space]

Example: Evaluate $\int_C F \cdot dr$ where C is the ellipse $4x^2 + 9y^2 = 1$ and $F(x,y) = \left((1+xy)e^{xy}, x^2 e^{xy}\right)$.

$$\int_C F \cdot dr = \int_C \nabla f \cdot dr = 0$$

w/ $f = xe^{xy}$

b/c C is a closed loop!

Questions?

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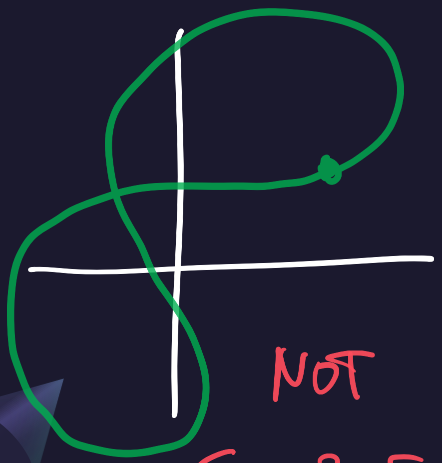
Green's Theorem

About Parametrized Curves

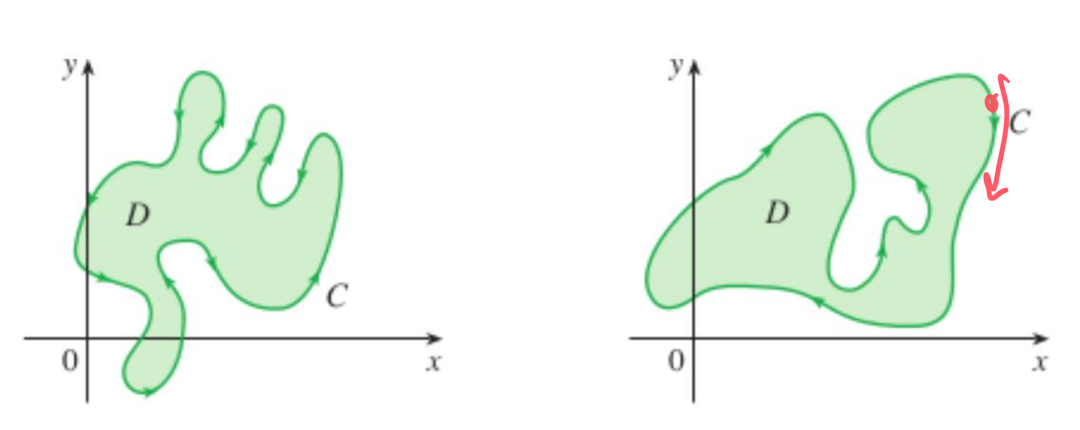
A parametrized curve is called closed or a loop if it starts and ends at the same point.

A closed curve is simple if its intersection with itself is at the start and the end, and nowhere else.

A simple closed curve on the plane is positively oriented if it goes counterclockwise. Otherwise it is negatively oriented.



NOT SIMPLE.



(a) Positive orientation

(b) Negative orientation

POS.

NEG.

Green's Theorem

Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C. If P and Q have continuous partial derivatives on an open region that contains D, then

$$\int_C F \cdot dr = \int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$F = (P, Q)$$

Example: Evaluate $\int_C x^4 dx + xy\, dy$ where C is the triangular curve along the segments from (0,0) to (1,0), from (1,0) to (0,1), and from (0,1) to (0,0).

[Extra Space]

Example: Evaluate $\int_C x^4 dx + xy\, dy$ where C is the triangular curve along the segments from (0,0) to (1,0), from (1,0) to (0,1), and from (0,1) to (0,0).

Green's Theorem

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Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C. If P and Q have continuous partial derivatives on an open region that contains D, then

$$\int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

Notes: Sometimes we write $\oint_C$ … to denote we are integrating along a closed curve.

If F = (P,Q) is conservative, then $F = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$ and therefore

$$\frac{\partial P}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial Q}{\partial x}$$

Example: Evaluate $\oint_C \left(3y - e^{\sin(x)}\right)dx + \left(7x + \sqrt{y^4+1}\right)dy$ where C is the circle of radius 3 centered at the origin.

[Extra Space]

Example: Evaluate $\oint_C \left(3y - e^{\sin(x)}\right)dx + \left(7x + \sqrt{y^4 + 1}\right)dy$ where C is the circle of radius 3 centered at the origin.

Green's Theorem

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$$

Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C. If P and Q have continuous partial derivatives on an open region that contains D, then

$$\int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

$$P(x,y) = 0 \qquad P(x,y) = -y \qquad P(x,y) = -\frac{1}{2}y$$

$$Q(x,y) = x \qquad Q(x,y) = 0 \qquad Q(x,y) = \frac{1}{2}x$$

$$A = \oint_C x\,dy = -\oint_C y\,dx = \frac{1}{2}\oint_C x\,dy - y\,dx$$

Example: Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

[Extra Space]

Example: Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Example: Find the area of the triangle with vertices (0,0), (2,0), (2,3).

[Extra Space]

Example: Find the area of the triangle with vertices (0,0), (2,0), (2,3).

Questions?

Thank you

Until next time.